

RAFFLES INSTITUTION
2021 Year 6 Preliminary Examination

Higher 2

BIOLOGY

Paper 1 Multiple Choice

9744/01

27th September 2021

1 hour

Additional Materials: Multiple Choice Answer Sheet

READ THESE INSTRUCTIONS FIRST

Write in soft pencil.

Do not use staples, paper clips, highlighters, glue or correction fluid.

Write your name and shade your Index Number on the Answer Sheet in the spaces provided unless this has been done for you.

There are **thirty** questions in this paper. Answer all questions. For each question there are four possible answers **A, B, C, and D**.

Choose the **one** you consider correct and record your choice in **soft pencil** on the separate Answer Sheet.

Read the instructions on the Answer Sheet very carefully.

Each correct answer will score one mark. A mark will not be deducted for a wrong answer.

Any rough working should be done in this booklet.

The use of an approved calculator is expected, where appropriate.

(Erase all mistakes completely. Do not bend or fold the OMR Answer Sheet).

This document consists of **22** printed pages.



1. A student carried out the Benedict's test on two different types of milk, X and Y.

A sample of each type of milk was heated to 100°C in a water-bath with Benedict's solution and the time taken for the first appearance of a colour change was recorded.

The results are shown in the table.

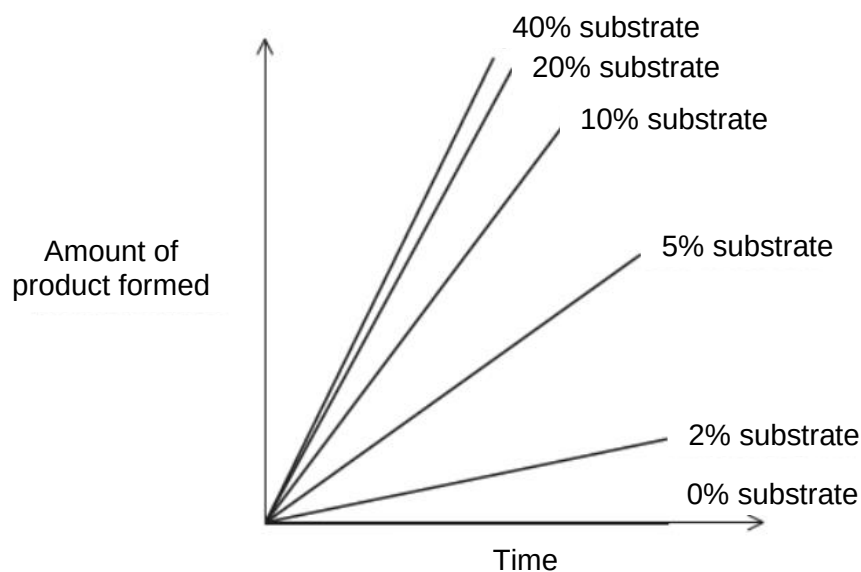
type of milk	time for first appearance of a colour change with Benedict's solution/s
X	13
Y	26

Which row shows the biological molecule the student detected in each sample of milk and the sample of milk with the highest concentration of this biological molecule?

	biological molecule present in each sample of milk	sample of milk with the highest concentration of this biological molecule
A	glucose	X
B	glucose	Y
C	reducing sugar	X
D	reducing sugar	Y

X shows almost immediate presence of reducing sugar in 13s. Whereas in Y, time was taken to break down complex sugars into reducing sugars with heat. We cannot conclude from this experiment what type of reducing sugar was present.

2. The graph shows the effect of changing the substrate concentration on the early stages of an enzyme-catalysed reaction.

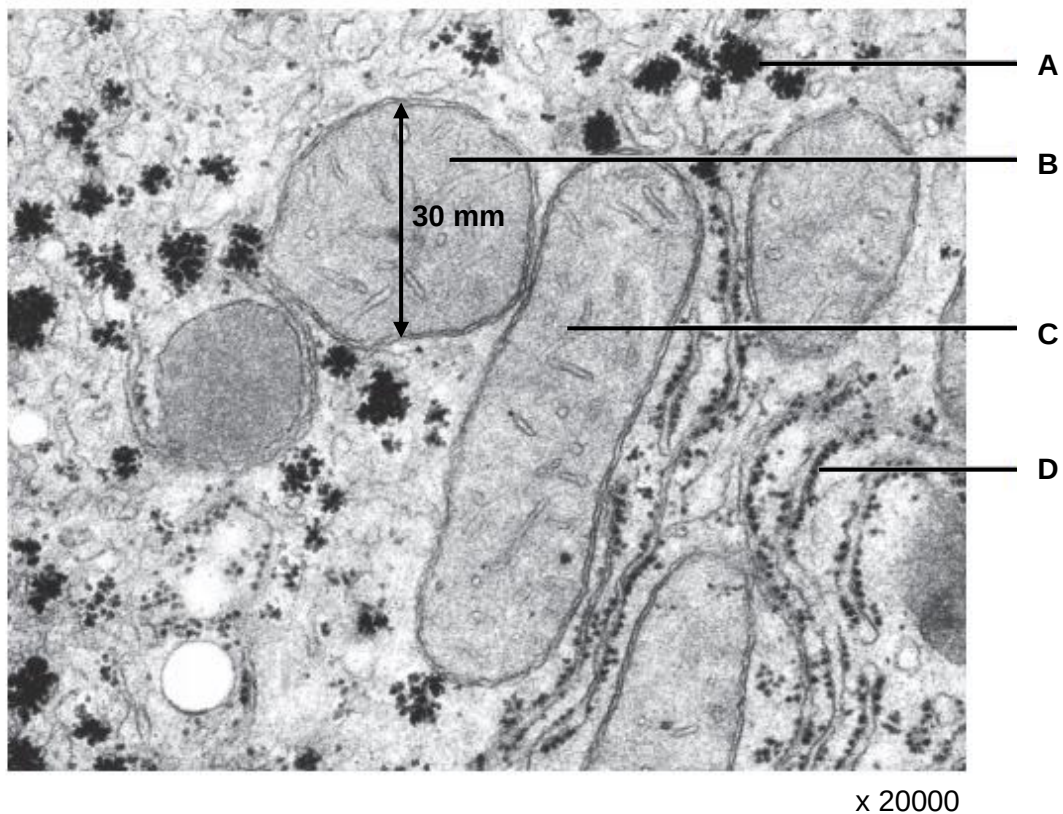


What can be interpreted about the rate of reaction from the graph?

The rate of reaction at each concentration of substrate can be determined by calculating the gradient of each line.

- A** The rate of reaction increases up to a point and then remains constant.
Not true, if this is indeed true. The gradient of the line for each 20% and 40 % should be the same
- B** The rate of reaction increases linearly with increasing substrate concentration.
Not true based on the gradient of the lines at each substrate concentration
- C** The rate of reaction increases non-linearly with increasing substrate concentration.
True based on the gradient of the lines at each substrate concentration
- D** The rate of reaction is not affected by any change in the substrate concentration.
Not true, if this is indeed true. The gradient of the line for all the substrate concentration will be the same.

3. The electron micrograph below shows some organelles and densely stained storage molecules in a mammalian liver cell.



Which of the following statements are true?

- 1 B and C are the same organelle.
True. Yes, they are. B is the cross section and C is the longitudinal section of 2 different mitochondria.
- 2 Glycogen phosphorylase catalyses the formation of A from glucose.
Not true. This is a liver cell. It stores, A, which is glycogen. Glycogen synthase catalyses the synthesis of glycogen from glucose, not glycogen phosphorylase. Glycogen phosphorylase catalyses the synthesis of glucose from glycogen.
- 3 The diameter of organelle B is 30mm, so its actual diameter is 1.5µm.
True

$$\text{Magnification} = \frac{\text{length of drawing or picture}}{\text{Actual length of specimen}}$$

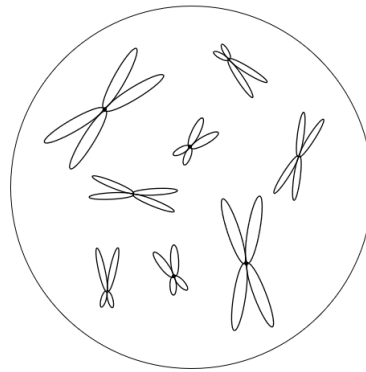
$$20000 = \frac{30 \text{ X } 1000}{\text{Actual length of specimen}}$$

$$\text{Actual length of specimen} = \frac{30000}{20000} = 1.5 \mu\text{m}$$
- 4 Amino acyl tRNA synthetase is synthesised at D.
Not true. Amino acyl tRNA synthetase is involved in amino acid activation that occurs in the cytosol. So it is synthesised on the free ribosomes in the cytosol, not the ribosomes on the RER that synthesise proteins meant for insertion into the cell membrane, packaging into vesicles and for secretion out of the cell.

- A 1 and 3 only
- B 2 and 3 only
- C 2 and 4 only
- D 1, 3 and 4 only

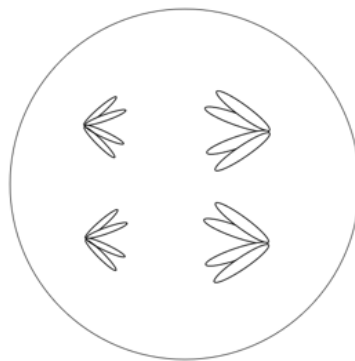
4. Which statement does not describe any of the membranes of the endomembrane system?
The endomembrane system includes the membranes of: plasma membrane; nuclear envelope; RER; SER; Golgi Apparatus; lysosomes; vesicles.
- A In antigen presenting cells, MHC proteins are embedded in the membrane after synthesis. *True. This refers to the plasma membrane.*
 - B The membrane may possess proteins that transport pyruvate into the organelle it surrounds. *False. This refers to pyruvate translocase which transports pyruvate from the cytosol to the mitochondria and is found on the inner mitochondrial membrane, which is not part of the endomembrane system.*
 - C Fluidity of the membrane allows for the secretion of proteins like insulin out of the cell and into the blood. *True. This refers to the plasma membrane and the release of insulin out of the cell via exocytosis.*
 - D Compartmentalisation by the membrane allows an organelle to maintain a lower pH compared to the rest of the cytosol. *True. This refers to the membrane of the lysosome which has a low pH compared to the cell cytosol, for the hydrolytic enzymes it contains to function.*

5. The diagram represents the nucleus of a cell $2n = 8$ in late prophase of mitosis.

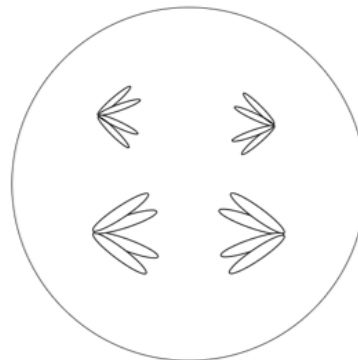


Which diagram represents a cell from the same species in anaphase II of meiosis?

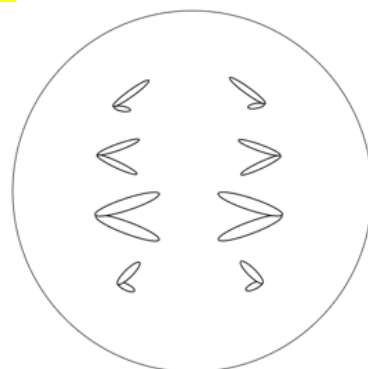
A



B



C



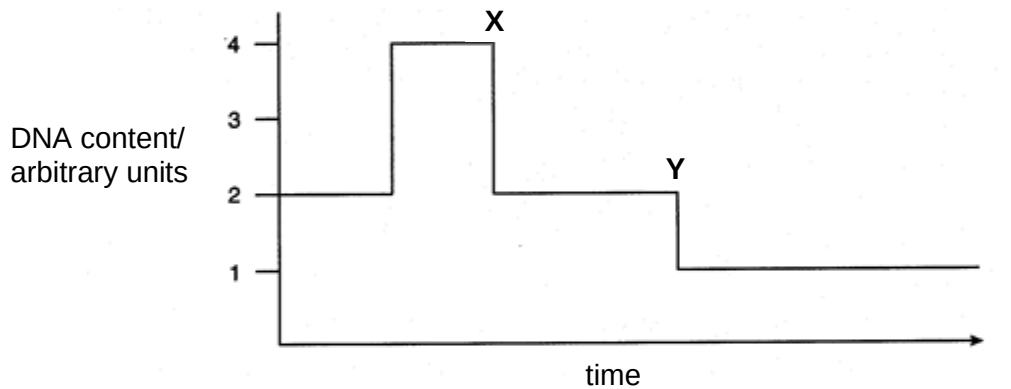
D



C shows that in anaphase II, centromere divide and the chromatids separate and form daughter chromosomes that are pulled to opposite poles.

A, B and D shows chromosomes with 2 chromatids being pulled to opposite poles which is not possible in anaphase II

6. The graph shows the amount of DNA present in the nuclei of cells undergoing division in a multicellular organism.



Which combination is correct?

DNA content decreases from 4 au to 2 au at X, this shows telophase I.
DNA content further decreases from 2 au to 1 au at Y, this show telophase II.

Answer is not A, because the question stem stated that “*The graph shows the amount of DNA present in the nuclei of cells*”

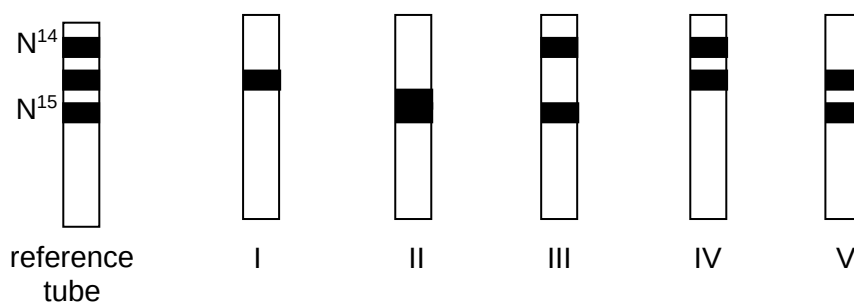
Ans could be A, if the questions stem stated that “*The graph shows the amount of DNA present in the cell*”

Be very careful with these subtle difference in description

	type of nuclear division	X	Y
A	meiosis	cytokinesis	cytokinesis
B	meiosis	telophase I	telophase II
C	mitosis	anaphase	telophase
D	mitosis	metaphase	telophase

7. Cells of the *Escherichia coli* bacterium were grown for many generations in a medium containing only the heavy isotope of nitrogen, ^{15}N . The cells were then transferred to a medium containing only ^{14}N and allowed to grow.

Samples of the bacteria were removed from the culture after the 1st round of replication and then again after the 2nd round of replication. The DNA from each sample was extracted and centrifuged at very high speed in a solution of caesium chloride.



The diagram above illustrates the DNA distribution in various centrifuge tubes, I, II, III, IV and V as proposed by a student.

Which of the above centrifuge tubes show semi-conservative DNA replication?

After 1st round of replication in ^{14}N medium: 100% ^{15}N - ^{14}N

After 2nd round of replication in ^{14}N medium: 50% ^{15}N - ^{14}N , 50% ^{14}N - ^{14}N

	tube after 1 st round of replication	tube after 2 nd round of replication
A	I	III
B	IV	III
C	I	IV
D	II	I

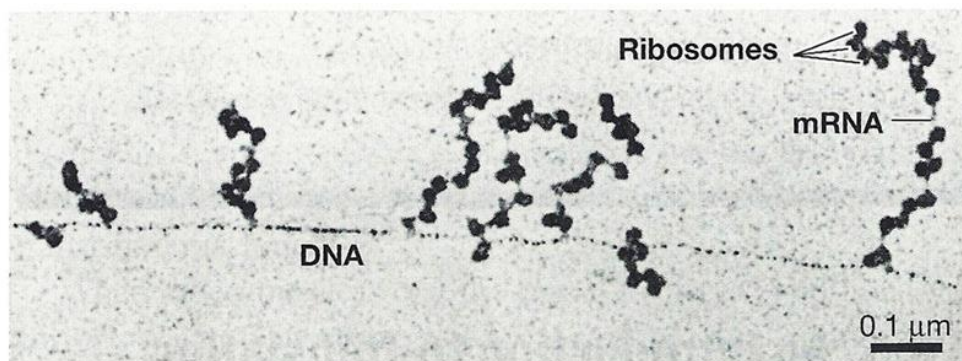
8. The mechanisms of action of five antibiotics are stated below.

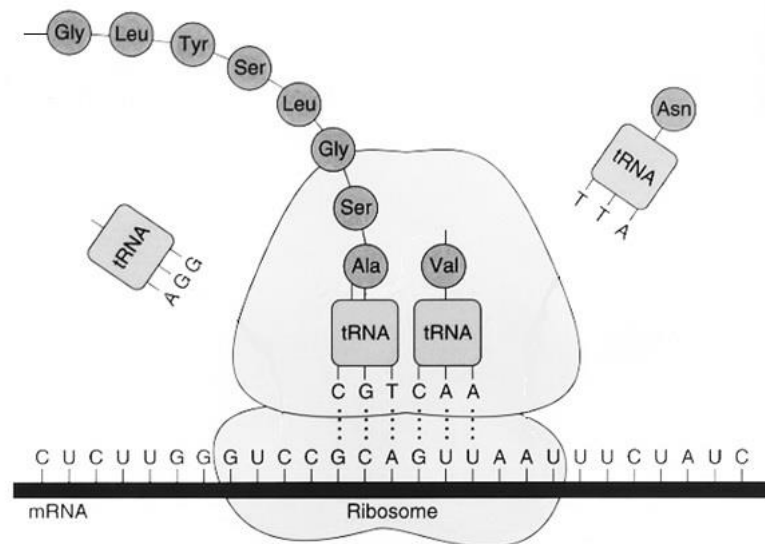
antibiotic	mechanism of action
tetracycline	binds to the 30S ribosomal subunit of a bacterium, blocking aminoacyl tRNA from binding to the A-site <i>Results in a truncated polypeptide/no translation.</i>
clindamycin	inhibits peptidyl transferase function <i>Results in a truncated polypeptide/no translation.</i>
rifampin	inhibits DNA-dependent RNA polymerase during transcription initiation <i>Results in no transcription/premature termination of the transcription process.</i>
quinolone	inhibit the ability of topoisomerase to join nucleotide fragments <i>Topoisomerase not being able to join back the strands when relieving the overwinding strain ahead of the replication fork results in a disrupted DNA template strand and no DNA replication.</i>
kanamycin	causes misreading of the mRNA and incorporation of incorrect amino acids <i>Results in proteins with abnormal function.</i>

Which row correctly matches the specified mechanisms of action of these antibiotics with the direct effects on a bacteria cell?

	polypeptide with abnormal function	truncated polypeptides	no transcription	no DNA replication
A	clindamycin	kanamycin	rifampin	tetracycline
B	kanamycin	tetracycline	quinolone	clindamycin
C	kanamycin	clindamycin	rifampin	quinolone
D	clindamycin	rifampin	quinolone	tetracycline

9. The first figure below shows gene expression occurring in a cell. The second figure contains more details on the process occurring at one of the ribosomes.



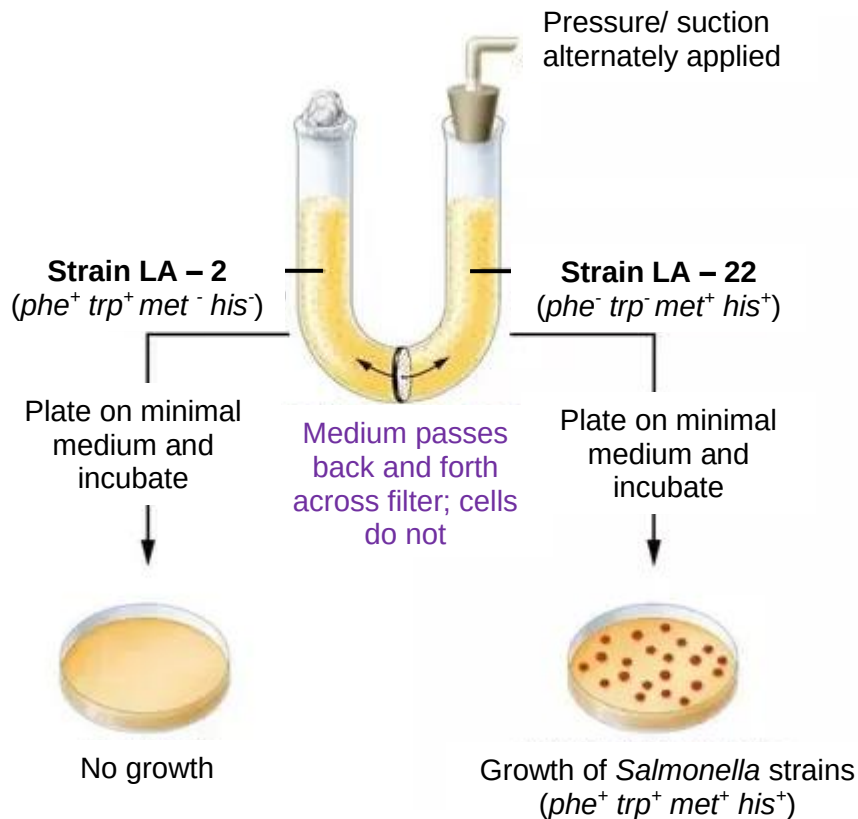


Which pair of statements is true about the gene expression process shown?

I	The ribosome is translocating from right to left. <i>False. In the second picture, the aminoacyl tRNA (holding the growing polypeptide chain) in the P site is to the left of the aminoacyl tRNA (with the incoming amino acid) in A site (left to right). Therefore the ribosome must be translocating from left to right so that the aminoacyl tRNAs can be eventually shifted in the direction of the E site.</i>
II	The information provided does not show degeneracy of the genetic code with respect to leucine. <i>True. Although there are two leucine amino acids in the polypeptide chain, the mRNA only shows the codon of one of them.</i>
III	These processes are occurring in the nucleus and cytoplasm of the cell. <i>False. Since transcription and translation are occurring simultaneously, these events are occurring in a prokaryotic cell where there is no nucleus.</i>
IV	The DNA is read from the 3' to 5' direction while the mRNA is read from the 5' to 3' direction. <i>True. Always the case for transcription and translation respectively.</i>

- A I and III only
- B I and IV only
- C II and III only
- D II and IV only**

10. In the experiment by Zinder and Lederberg in 1952, investigating genetic exchange in *Salmonella* strains was demonstrated using a U-tube set up with a filter separating the two arms as shown in the figure below.



Strain LA-2 is able to synthesise phenylalanine and tryptophan but not methionine and histidine. Strain LA-22 is able to synthesise methionine and histidine but not phenylalanine and tryptophan.

Which of the following conclusions can be drawn from the results?

- I The method of genetic transfer must be transduction.
False. Because the question doesn't provide information about the pore size of the filter or whether nucleases are added to the medium, so we cannot rule out transformation as fragments of DNA from lysed cells could also pass through the filter.
 - II Direct contact between cells is not required for the transfer of genes.
*True: the filter prevents cells from having direct contact, yet LA-22 managed to obtain the *phe⁺* and *trp⁺* alleles from LA-2.*
 - III LA-2 is resistant to phage infection.
*False: Since LA-2 is the donor of *phe⁺* and *trp⁺* alleles the phage must have infected it first in order to obtain those alleles, proving that it is not resistant to phage infection.*
 - IV LA-2 is the donor of the *phe⁺* and *trp⁺* genes.
True.
- A** IV only
- B** I and II only
- C** II, and IV only
- D** I, II and III only

11. Inducible genes code for the synthesis of enzymes involved in catabolic pathways.

Which mechanism in bacteria can be used to control the activity of inducible genes?

- A Excess product of the catabolic pathway binds to the repressor to inactivate it.
- B Excess substrate of the catabolic pathway binds to the operator, blocking RNA polymerase from binding to the promoter.
- C** Excess substrate of the catabolic pathway binds to the repressor to inactivate it.
- D Excess product of the catabolic pathway binds to the repressor to activate it.

Comments:

Inducible operon is by default off (repressor active) and can be switched on (repressor made inactive) in the presence of the substrate (so that its protein products can break it down).

12. Which of the following statement/s on viruses is/are false?

- 1 The HIV polyprotein is synthesised from a polycistronic mRNA template.
- 2 Influenza B undergoes genetic drift as there is an accumulation of mutations in the haemagglutinin gene with time.
- 3 T4 phage is the agent involved in specialised transduction.
- 4 Replication cycles of both HIV and lambda phage have a latent phase.

- A 3 only
- B 4 only
- C 1 and 2 only
- D** 1, 2 and 3 only

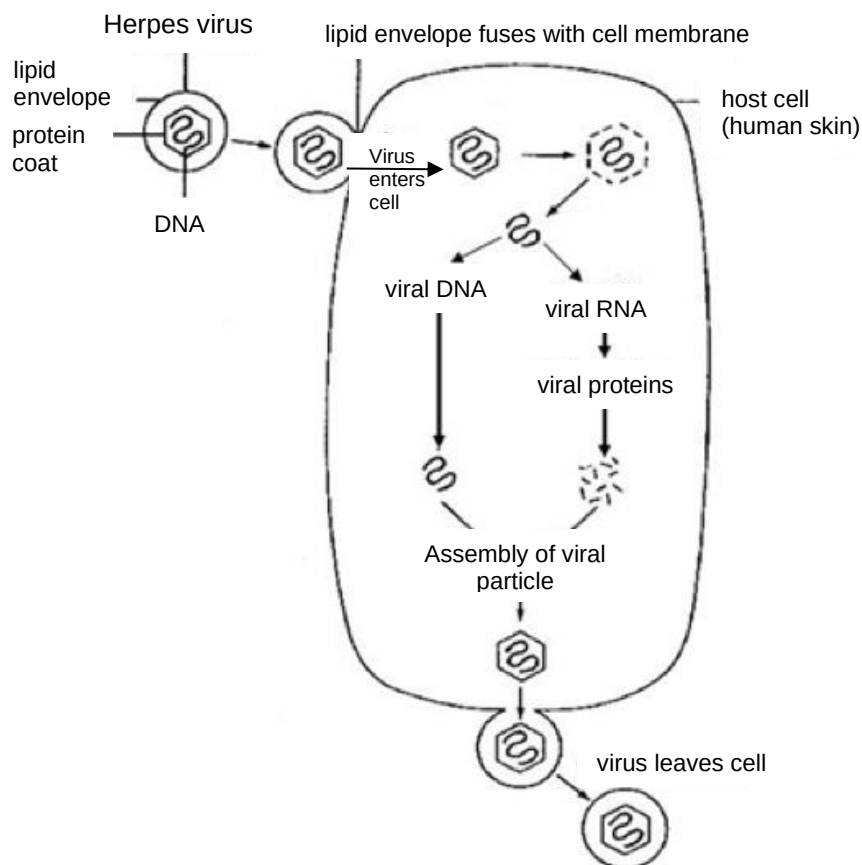
Comments:

*1 The HIV polyprotein is synthesised from a ~~polycistronic~~ **monocistronic** mRNA template. HIV infects eukaryotic cells which have monocistronic mRNA.*

*2. Influenza B undergoes ~~genetic~~ **antigenic** drift as there is an accumulation of mutations in the haemagglutinin gene with time.*

*3. ~~T4~~ **lambda** phage is the agent involved in specialised transduction.*

13. The diagram below shows the replication cycle of the herpes virus which causes cold sores in the mouth.



Which of the following rows represents the herpes virus replication cycle?

	penetration	enzyme involved in genome replication	release
A	endocytosis	RNA dependant RNA polymerase	exocytosis
B	fusion	DNA dependant RNA polymerase	budding
C	endocytosis	reverse transcriptase	exocytosis
D	fusion	DNA polymerase	budding

Comments:

The genome of the virus is labelled DNA hence genome replication has to be by DNA polymerase. Entry by fusions and release by budding can be seen in the diagram.

14. Which of the following statement(s) is/are incorrect regarding telomeres?

- 1 Telomeres prevent the end-replication problem.

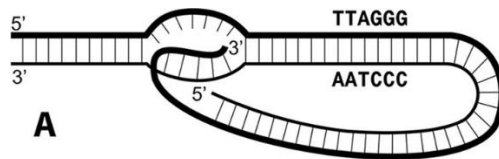
Incorrect. They do not prevent the end-replication problem from happening.

- 2 Telomeres prevent degradation of chromosomes by nucleases.

Incorrect. The degradation of chromosomes will still occur, but in the presence of telomeres, the genes are not eroded.

- 3 Telomeres prevent the ends of chromosomes from fusing via complementary base pairing.

Correct. 2 chromosomes can fuse together by complementary base pairing of each 3' overhang. Telomeres prevent this from happening by stabilising the ends (by forming a loop, with the ss portion displacing a portion of the ds region by complementary base pairing with it).



- 4 The single stranded portion of the telomere serves as a template for extension via complementary base pairing.

Incorrect. The template for extension of telomeres is provided by the RNA subunit of the telomerase enzyme.

A 1 only

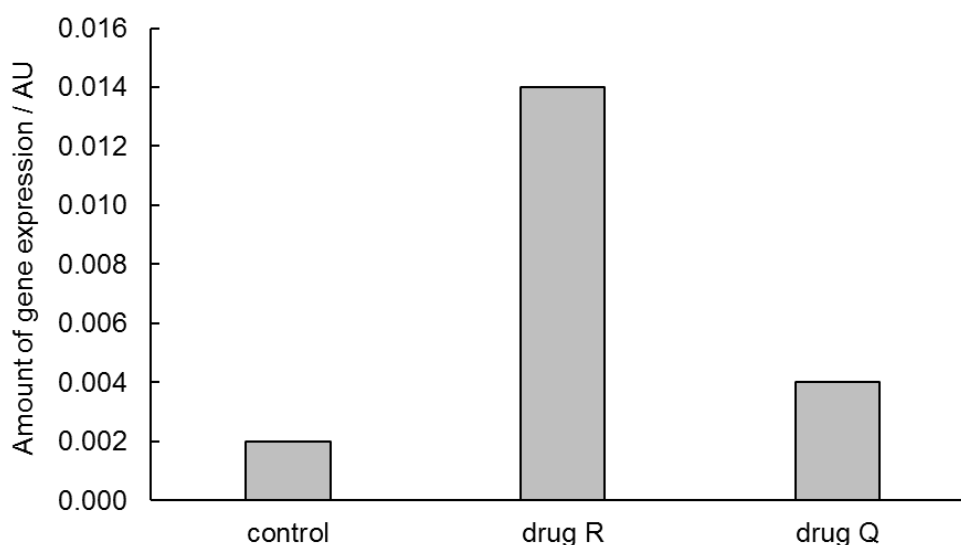
B 1 and 4 only

C 3 and 4 only

D 1, 2 and 4 only

15. Drug R is a DNA methyltransferase inhibitor and drug Q is a histone deacetylase inhibitor. An experiment was carried out to investigate the effects of drug R and Q on expression of a gene.

The graph shows the experimental results.



Based on the results shown which of the following could be true?

Comments:

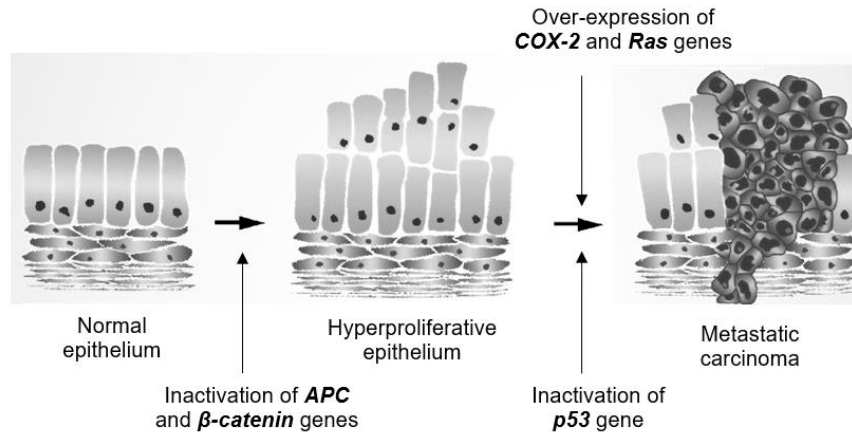
The enzymes DNA methyltransferase and histone deacetylase will result in gene silencing.

The drugs R and Q (which are their inhibitors respectively) should result higher level of gene expression (compared to the absence of inhibitor). The higher the amount of gene expression, the more effective the inhibitor.

- 1 The control experiment shows the level of gene expression in the absence of both drugs.
True. Control experiment shows the lowest level of gene expression, which is possibly due to the absence of both drugs. Also, usually the control experiment is the experiment where there is the absence of the treatment (inhibitory drug in this case)
- 2 Drug Q controls modification of histones, resulting in weaker binding of histones to DNA.
True. Drug Q in a histone deacetylase inhibitor and therefore prevents removal of acetyl groups. The presence of acetyl groups on lysine residues results in weaker electrostatic attraction between DNA and histones.
- 3 Drug Q increases gene expression by increasing accessibility of RNA polymerase to the promoter.
True. See above.
- 4 Drug R increases gene expression by preventing DNA methylation.
True. DNA methyltransferase adds methyl groups. Drug R prevents that.
- 5 Drug R is less effective at inhibiting gene expression than drug Q.
Not true. The higher the amount of gene expression, the more effective the inhibitor (see above explanation)

- A** 5 only
B 2, 3 and 4 only
C 1, 2, 3 and 4 only
D All of the above

16. The diagram below illustrates the development of colorectal cancers.



Based on the diagram as well as your understanding of how cancer develops, which of these statements can be inferred from this multistep model of carcinogenesis?

- 1 Cells whose *APC* and *β-catenin* genes are inactivated have lost contact inhibition and can form a tumour mass.
True. It can be seen from the figure.
 - 2 *APC* and *β-catenin* genes are tumour suppressor genes.
True, since loss of function led to tumour formation.
 - 3 High levels of *Ras* protein are produced only when both copies of *Ras* gene are mutated.
Not true. As Ras is a proto-oncogene (can be identified from the figure since over-expression led to metastasis). Proto-oncogene only require a mutation in one copy of the allele to result in a gain in function
 - 4 Two copies of normal *p53* alleles must be present to inhibit cell division.
Not true. It can be inferred from the figure that p53 is a tumour suppressor gene. Only one normal copy is needed for normal function.
 - 5 Gain-of-function mutation in *COX-2* gene increases the chances of metastatic carcinoma.
True. Can be inferred directly from the figure.
- A** 1 and 3 only
B 2, 3 and 4 only
C 1, 2 and 5 only
D 2, 3 and 5 only

17. Which of the following statements are correct?

- 1 Chlorophyll b is an accessory pigment that is involved in the resonance transfer of electrons from one pigment molecule to the next in the light harvesting complex of both photosystems I and II.

Incorrect. It is the resonance transfer of energy not electrons.

- 2 ATP synthase is located at the thylakoid membrane and the membrane of the intergranal lamella. *Correct.*

- 3 When reduced, NADP can donate electrons to the electron carriers in the electron transport chain on the thylakoid membrane.

Incorrect. It reduces GP to G3P in the Calvin cycle in the stroma.

- 4 The electrons released during the photolysis of water can be used to reduce NADP during non-cyclic photophosphorylation. *Correct. Electrons from the photolysis of water fills the electron hole in the reaction centre of PS II. The path it subsequently takes is as follows: primary electron acceptor → 1st ETC → PS I → 2nd ETC → NADP*

A 1 and 2 only

B 2 and 4 only

C 3 and 4 only

D 1 and 3 only

18. Some information about the respiratory enzymes that catalyse the reactions is shown in the table below.

enzyme	information
fructose 1,6-bisphosphate aldolase	- four identical subunits - changes to any one of the subunits means that the enzyme cannot function
hexokinase	- one subunit - active site changes shape to enclose the reactants
phosphofructokinase	- four identical subunits - has allosteric sites in addition to an active site
phosphoglucose isomerase	- two identical subunits - has a cytokine function when secreted into the external medium
pyruvate kinase	- four identical subunits - ATP acts as an inhibitor to regulate glycolysis
triosephosphate isomerase	- two identical subunits - each subunit has 14 alpha helices and 8 beta-pleated sheets

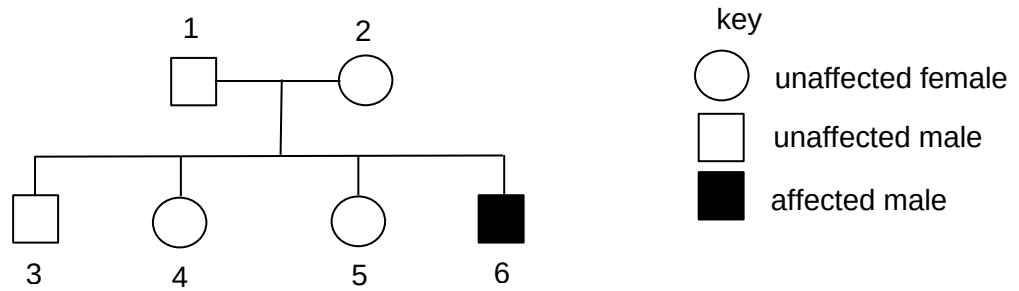
A student made the following deductions using the information provided in the table:

- Phosphoglucose isomerase, when secreted, can have a non-catalytic role. *Correct. Cytokines are not enzymes*
- Only three of the six enzymes display quaternary protein structure. *Incorrect. 5 enzymes display quaternary structure. All except hexokinase.*
- The active site of phosphofructokinase will change shape to allow the enzyme to regulate the rate of glycolysis. *Correct. PFK is an allosteric enzyme involved in glycolysis. Inhibitors/Activators can bind to the allosteric site and change the shape of the active site.*
- Each enzyme is coded for by one gene. *Correct. All the subunits are identical.*
- The reaction catalysed by hexokinase is an induced-fit mechanism. *Correct. The shape of the active site changes to enclose the reactants/substrates.*

How many of the student's deductions are correct?

- A 1 B 2 C 3 **D 4**

19. The diagram shows the inheritance of haemophilia in a family.



If individual 4 married a normal male, what is the probability that their first child would suffer from haemophilia?

- A 0 B 0.125 C 0.25 D 0.5

Comments:

Expectation to know that haemophilia is a sex-linked recessive condition.

If 6 has the condition, genotype is X^hY and so have inherited X^h allele from mother 2.

Since 2 is normal phenotype, her genotype is X^HX^h .

Probability of 4 inheriting the X^h allele from 2 is 0.5.

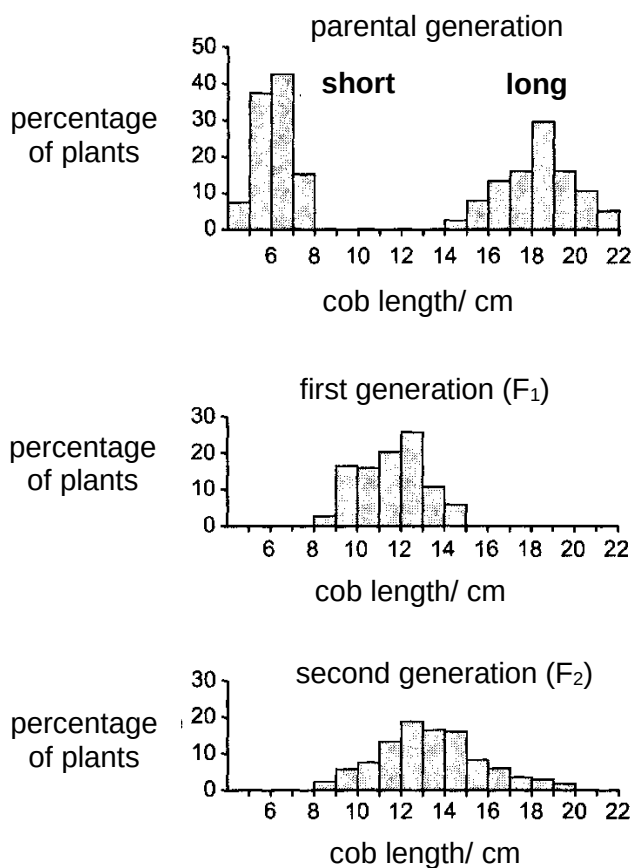
If 4 has genotype X^HX^h and marries a normal male X^HY , then offspring genotypes will be:

X^HX^H , X^HX^h , X^HY and X^hY .

Probability of offspring suffering from haemophilia is 0.25.

Total probability = $0.5 \times 0.25 = 0.125$

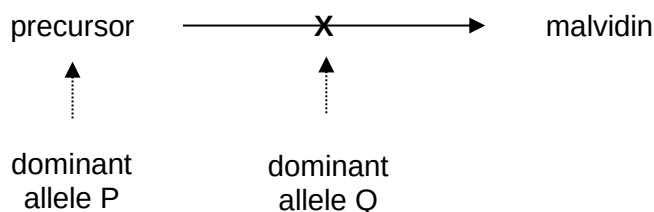
20. The graphs show the distribution of measurements of cob length of corn kernel in three corn generations.



Which of the following is evidence that corn cob length is not controlled by a single pair of alleles showing incomplete dominance?

- A** The corn plants do not fall into distinctive, clear-cut groups.
Range of phenotypes indicate polygenic inheritance, i.e. controlled by multiple genes. But does not explain incomplete dominance.
- B** The phenotypic variation in each generation shows the influence of environmental factors.
Range of phenotypes for parental and F₁ generations where there is only one genotype but a narrow range of phenotypes indicate the effect of the environment on the phenotypes. But does not explain incomplete dominance.
- C** The F₂ generation does not occur in a 1: 2: 1 ratio.
Range of phenotypes for F₂ generation indicate multiple genes. Also lack of 1:2:1 ratio when two F₁ heterozygotes are crossed indicated not incomplete dominance.
- D** The F₁ generation is of intermediate corn cob length.
Does not provide evidence for single pair of alleles showing incomplete dominance.

21. In the *Primula* plant, production of the pigment malvidin creates blue flowers. Synthesis of malvidin is controlled by a dominant allele **P** at the **P/p** gene locus which codes for a colourless precursor substance. However, the conversion of the precursor substance to malvidin can be suppressed by another dominant allele **Q** at the gene locus **Q/q** as shown in the biochemical pathway below. Flowers with the absence of malvidin will appear white.



When two plants with white flowers are crossed, the offspring generation has 44 plants with white flowers and 10 plants with blue flowers.

Which of the following statements is not true?

- A** The probability of obtaining a pure-breeding offspring at both loci with white flowers is 1 of 9.
False. PPQQ, ppQQ, ppqq are all pure-breeding offspring with white flowers.
- B** The deviation from the Mendelian ratio (9:3:3:1) is due to gene interaction in which the expression of gene Q masks the phenotypic expression of gene P.
True, due to epistatic interaction where gene product of Q masks the expression of gene P.
- C** The genotypes of the parents must be PpQq x PpQq.
True. To get the modified 9:3:3:1 ratio, both parents have to be heterozygous.
- D** The genotypes PPQQ, PPQq, PpQq and ppQq produce the same phenotype.
True. P_Q_ or ppQ_ will both code for white flowers.

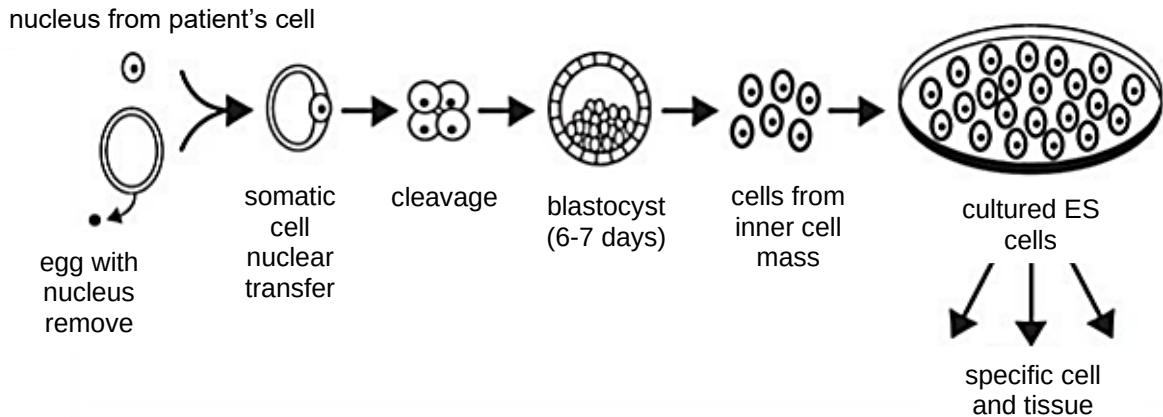
Comments:

*44 white flowers : 10 blue flowers
 = 13 : 3*

9 P_Q_ : 3 ppQ_ : 1 ppqq : 3 P_qq
 └──────────┘
13 white flowers : 3 blue flowers

22. The use of embryonic stem (ES) cells has encountered resistance on ethical grounds. To overcome this, researchers have come up with a way of developing embryonic stem cells from the cells of affected persons.

The figure below shows the theoretical process of development of these embryonic stem cells. The differentiated cell types would be introduced into the patient where needed.

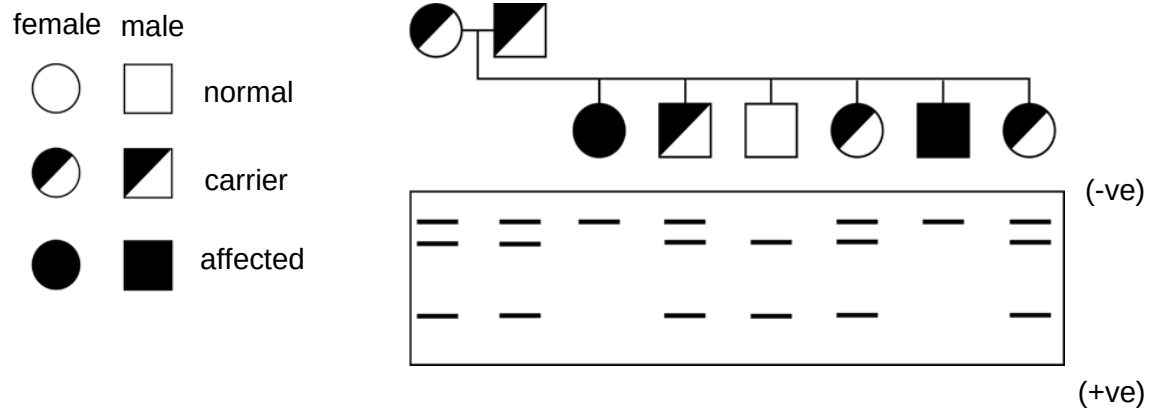


Which of the following option(s) is/are true?

- I The affected persons will not experience much of an immune response.
True. Since the nucleus is from the patient himself, the cells will express proteins that are the patient's and will not trigger an immune response.
- II Issues of informed consent are no longer relevant since the embryo is not donated.
False. Informed consent is an issue when it comes to egg, embryo and tissue donation. In the above process, eggs supplied by donors are still needed.
- III No embryo is destroyed in the process.
False. The process involves obtaining cells from the inner cell mass of the blastocyst, which would destroy the embryo that had developed till that point.
- IV The argument that stem cell research violates the sanctity of life is not relevant here.
False. Embryo destruction occurs in this process. Embryos are considered by some to have the status of a human being and so the argument would still apply.

- A** I only
- B** I and II only
- C** II and III only
- D** All of the above

23. In order to understand the inheritance of disease X, DNA of members of a family were subject to polymerase chain reaction, gel electrophoresis and Southern blotting. The diagram below shows the autoradiograph obtained.



The disease must be an autosomal recessive disorder since:

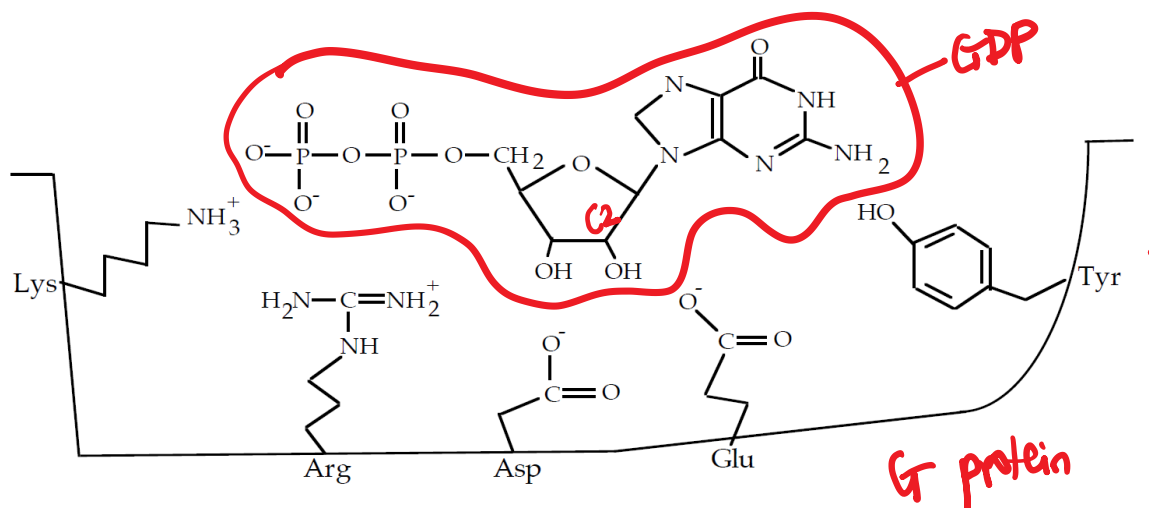
- 1. there are carriers with the normal phenotype but the disease allele.*
- 2. males can be a carrier too.*

Which of the following statements can be concluded from the autoradiograph?

- I The recessive allele is missing a restriction site.
True. Homozygous recessive have only 1 band while homozygous dominant individuals have 2 bands. A missing restriction site would explain the difference in the number of fragments.
- II The binding site of the radioactive probe does not contain the restriction site.
False. It likely does, which is why the same radioactive probe picks up fragments of 3 different sizes. Otherwise it would pick up only 2.
- III The disease is sex-linked recessive.
False. If this were the case, one of the genders cannot be carriers since they would only have one allele for the gene.
- IV The nitrocellulose membrane was washed after nucleic acid hybridisation.
True. If the membrane had not been washed, the entire membrane would have turned black and it would be difficult to observe clear bands as shown in the figure.

- A I and III only
- B I and IV only**
- C II and III only
- D II and IV only

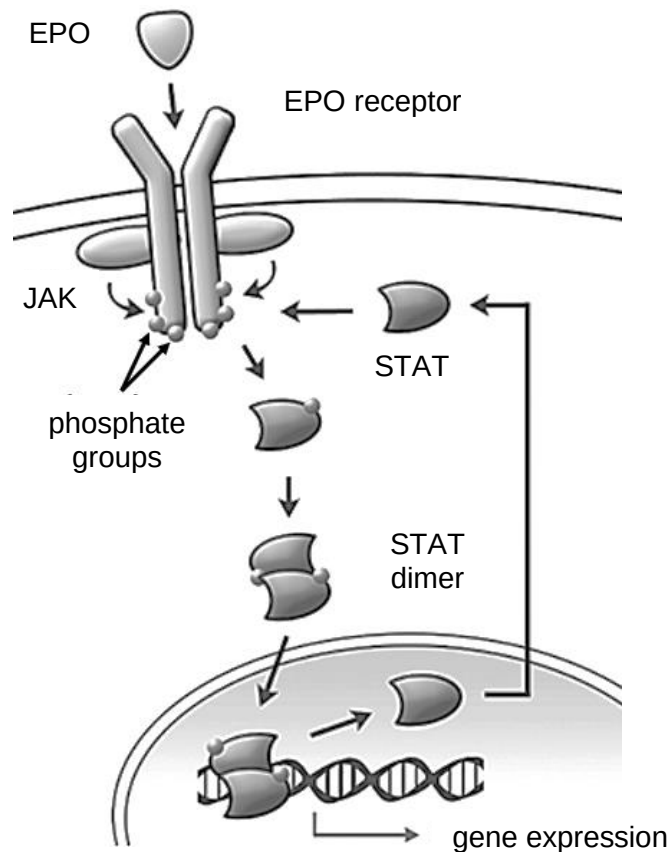
24. The figure below shows GDP in the binding pocket of a G protein.



Which of the following statement(s) is/are correct?

- GDP is attached to G protein because of complementary shape and charge.
- I *Correct. GDP fits into the binding site of the G protein by complementary shape and charge.*
- II Phosphorylation of GDP to GTP makes the binding to G protein stronger, activating the G protein in the process.
*Incorrect. GDP is **exchanged** with GTP to activate the G protein. GDP is **not** phosphorylated.*
- GDP is a deoxyribonucleoside diphosphate with guanine as the nitrogenous base.
- III *Incorrect. C2 on the pentose sugar of GDP has an OH group. Hence it is a ribonucleotide not a deoxyribonucleotide.*
- The binding of GDP to the G protein is via weak interactions.
- IV *Correct. To facilitate the exchange between GDP and GTP, the binding has to be reversible. Hence it is held in place via weak interactions.*
- A I only
- B I and IV only**
- C II and III only
- D All of the above

25. The diagram shows the JAK-STAT cell signalling pathway.



Which of the following statements are correct?

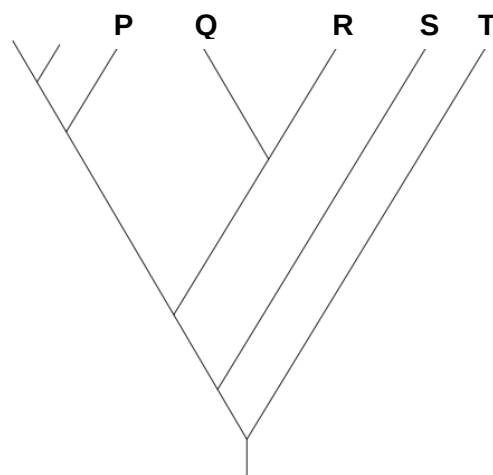
- 1 EPO is a large and non-polar lipid hormone and hence it needs a cell surface receptor to cross the membrane.
Incorrect. Non-polar molecules are able to pass through the hydrophobic core of the membrane as they are lipid soluble. They do not need a transmembrane receptor. Furthermore, EPO is an extracellular signal and does not need to enter the cell.
 - 2 Phosphorylation of STAT causes them to dimerize.
Correct. As seen in the diagram, once a phosphate group is added to each STAT subunit, they bind to each other to form the STAT dimer.
 - 3 Gene expression is terminated when phosphatases remove phosphate groups from STAT dimers.
Correct. When the phosphate group is removed, the STAT dimer dissociates from the DNA and returns to the cytoplasm. Gene expression will be terminated.
 - 4 Signal amplification occurs as JAK phosphorylates multiple tyrosine residues on the EPO receptor.
Incorrect. Although multiple tyrosine residues are phosphorylated, it is still 1 JAK activating 1 EPO receptor.
- A 1 and 3 only
B 2 and 3 only
 C 2 and 4 only
 D 2, 3 and 4 only

26. Information about two species of penguin and their habitats is shown in the table below.

name of penguin species	photograph of penguin	mean adult body mass/kg	mean adult height/cm	endemic to	temperature range/ °C
Emperor penguin (<i>Apenodytes forsteri</i>)		34.0	115.0	Antarctica	-60 to -28
Galapagos penguin (<i>Spheniscus mendiculus</i>)		2.2	50.0	Galapagos Islands	19 to 31

Molecular phylogeny suggests that the emperor penguin split off from a branch which led to the evolution of all other living penguin species. This split happened 40 million years ago.

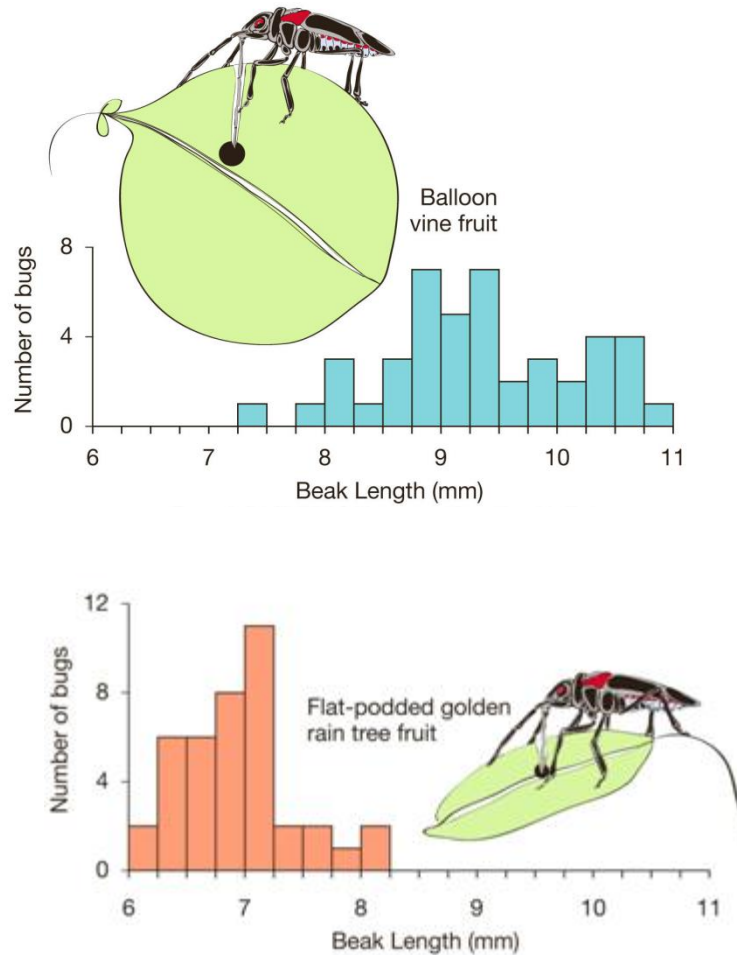
The diagram on the right indicates the evolutionary relationships between the main species of penguins. The letter Q indicates the position of the Galapagos penguin.



Based on the information given above, which of the following is correct?

	position of the emperor penguin on the diagram	position of the species most closely related to the Galapagos penguins on the diagram	anatomical adaptation of the emperor penguin to its habitat
A	S	P	having a large surface area to volume ratio to reduce heat loss
B	T <i>Since the emperor penguin split off from a branch which led to the evolution of all other living penguin species.</i>	R <i>Since Q (the Galapagos penguins) share a most recent common ancestor with R, they are the most closely related</i>	having a small surface area to volume ratio to reduce heat loss <i>The mean adult body mass/kg of the emperor is much larger than the Galapagos penguin. Hence is has a much larger surface area to volume ratio compared to the Galapagos penguin.</i>
C	R	S	reduced resting metabolic rate
D	P	R	thick layer of fat

27. Soapberry bugs in Florida originally fed on the native balloon vine, using their sharp beaks to penetrate the fruits to reach the seeds. In the 1920s, the flat-podded golden rain tree was introduced from Asia to mainland central Florida. This has thinner-skinned fruit, and correspondingly, soapberry bugs evolved shorter beaks after they switched completely to feed on this new host plant.



What best describes the type of evolutionary process depicted above?

The Soapberry bugs are undergoing

- A genetic drift
- B stabilising selection
- C disruptive selection
- D** directional selection

Comments:

As the frequency alleles the beak length in the population shifted from those that code for long beak length to those that code for short beak lengths after the 1920s. Directional selection is a mode of natural selection in which an extreme phenotype is favoured over other phenotypes, causing the allele frequency to shift over time in the direction of that phenotype.

28. Which statements about the use of antibiotics could cause an increase in antibiotic-resistant bacteria?

1 Patients do not always complete the course of antibiotics.

Not completing the course of antibiotics would mean that there could be too many bacteria in the body for the immune system to get rid off. These remaining bacteria could mutate and become antibiotic resistant.

2 Patients keep unused antibiotics from previous prescriptions and take them at a later date in smaller doses than prescribed.

Taking smaller doses of antibiotics is similar to not completing the course of antibiotics. It will not be sufficient to get rid of all the bacteria and the remaining bacteria could mutate and become antibiotic resistant. (if the bacteria had not already become resistant from the patient not completing the course of antibiotics from the previous prescription for the previous infection.)

3 Patients do not take antibiotics.

The use of antibiotics is the selection pressure. So as long as antibiotics are not consumed there will be no selection for antibiotic resistance strains.

4 Antibiotics are used in farming to prevent infections in livestock.

Using antibiotics consistently in farming as a preventative measure will select for antibiotic resistant strains and the susceptible ones will be killed by the antibiotics.

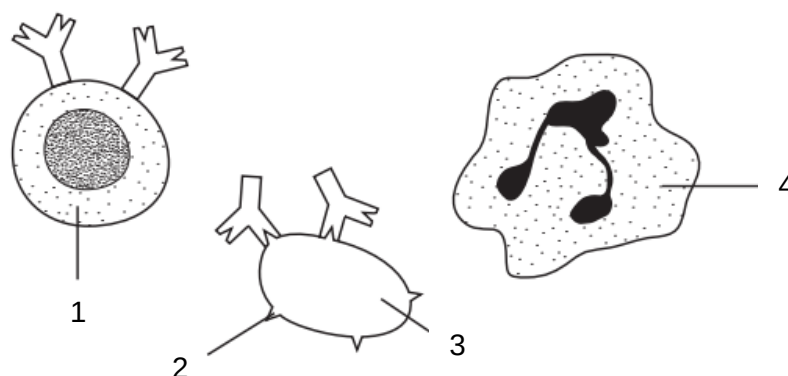
A 1, and 2 only

B 1 and 3 only

C 1, 2 and 4 only

D 3 and 4 only

29. The diagram shows different elements involved in the immune response in humans.



What are the parts labeled 1 to 4?

	1	2	3	4
A	Lymphocyte <i>Probably a naïve B lymphocyte since it has BCR</i>	Antigen <i>Looks like an epitope on a antigen</i>	Pathogen <i>Looks like a bacteria</i>	Phagocyte <i>Looks lik a neutrophil because of the multi-lobed nucleus</i>
B	phagocyte	antibody	pathogen	lymphocyte
C	phagocyte	antigen	lymphocyte	pathogen
D	pathogen	antigen	lymphocyte	phagocyte

30. What is the difference between T and B lymphocytes?

	T lymphocytes	B lymphocytes
A	do not form plasma cells	form plasma cells that secrete antibodies
B	does not stimulate macrophages to carry out phagocytosis <i>False statement</i>	stimulate macrophages to carry out phagocytosis <i>False statement</i>
C	formed from cells in the thymus <i>T lymphocytes are formed from cells in the bone marrow, then mature at thymus. So wrong comparison</i>	formed from cells in the bone marrow
D	produce memory cell	do not form memory cells <i>Both produce memory cells. So wrong comparison</i>

End of Paper

Answers:

1.	C	7.	C	13.	D	19.	B	25.	B
2.	C	8.	C	14.	D	20.	C	26.	B
3.	A	9.	D	15.	C	21.	A	27.	D
4.	B	10.	C	16.	C	22.	A	28.	C
5.	C	11.	C	17.	B	23.	B	29.	A
6.	B	12.	D	18.	D	24.	B	30.	A